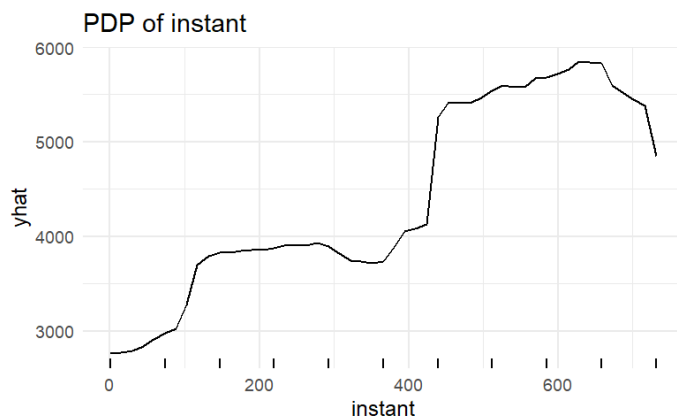


XAI: Model-agnostic methods

QUESTION: Analyse the influence of days since 2011, temperature, humidity and wind speed on the predicted bike counts.

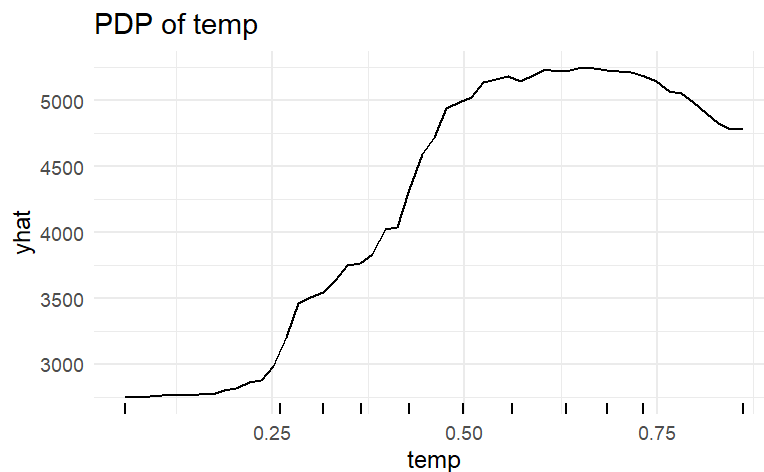
We are going to analyze the factors that cause the increase or decrease in bicycle rentals, specifically the passage of days, temperature, humidity, and wind speed.

Influence of instant (Days since 2011)



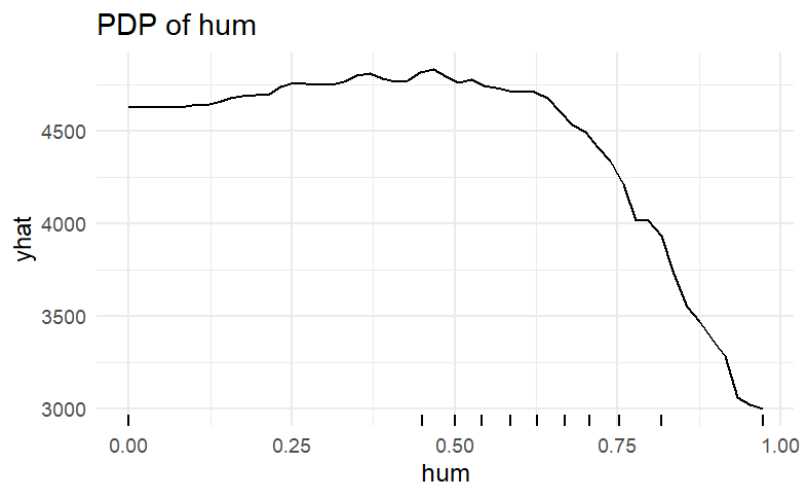
In the graph, we observe an increase in bicycle usage from instant 1 (2011-01-01) to instant 731 (2012-12-31). Specifically, the greatest increases in bicycle usage occur around instants 100 and 450, which correspond to the month of April, during springtime, while usage decreases during the Christmas holiday season. Overall, we can see that throughout these two years, the number of rented bicycles increased, making them more popular.

Influence of Temperature



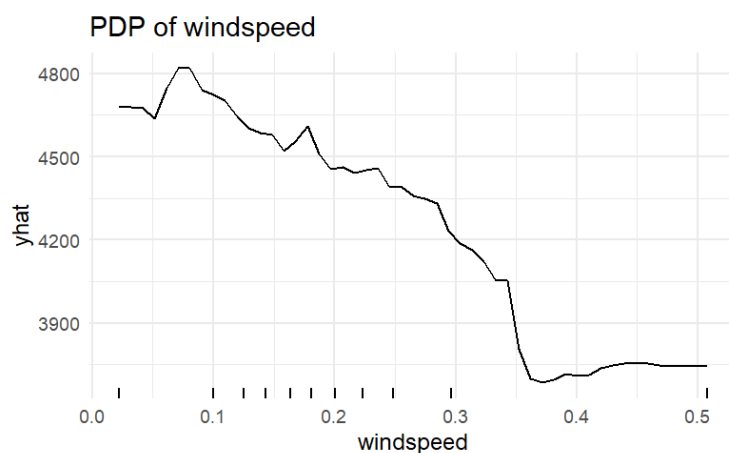
In this case, we observe that an increase in temperature leads to an increase in bicycle rentals. However, when temperatures become very high, the curve starts to decrease. These conclusions can be related to the previous ones, since lower temperatures coincide with winter and Christmas, resulting in very few bicycle rentals. On the other hand, as temperatures rise, we reach typical summer conditions, where the maximum number of rentals is observed (possibly because people have more free time for leisure activities during holidays, especially children and teenagers).

Influence of Humidity



We observe that humidity has a negative effect on bicycle rentals. Specifically, when humidity reaches approximately the 60% threshold, the steepest decline occurs. This can be related to the previous two graphs, since the highest humidity levels usually occur during colder and rainier months such as November and December, reinforcing the drop in demand during those periods.

Influence of windspeed

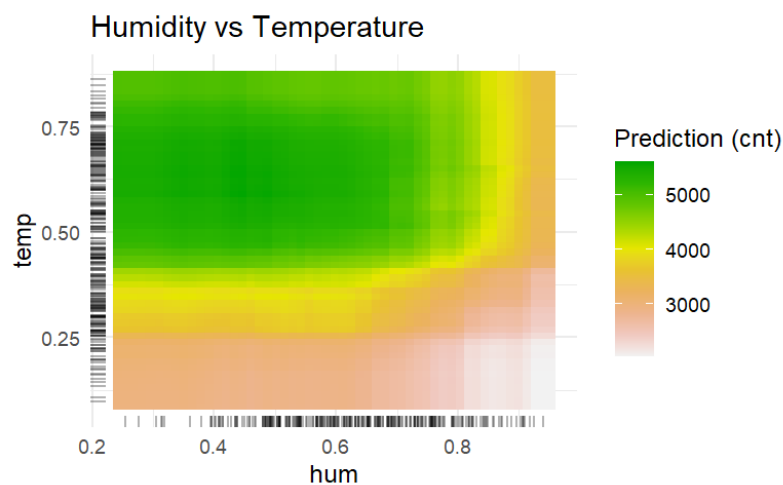


This graph is more difficult to connect with the others, but its individual analysis is clear: wind speed has a negative effect. Although small fluctuations can be observed in the data (such as slight changes between 0.05 and 0.17), they are not significant because the

differences in wind speed are small. The important aspect is the overall trend: as wind speed increases, bicycle rentals decrease, probably due to the discomfort or danger of cycling under strong wind conditions.

EXERCISE: Generate a 2D Partial Dependency Plot with humidity and temperature to predict the number of bikes rented depending on those parameters.

Previously, we tried to find the relationship between the number of bicycles and different variables individually. However, this time we will create a combined graph using humidity and temperature to see how they affect each other.



By looking at the graph, we realize that if the temperature is low (below 0.25), regardless of the humidity level, there will always be few rentals. This tells us that cold weather is the main factor limiting bicycle usage.

In general, we observe that the highest number of bicycle rentals occurs when temperatures are higher (between 0.5 and 1) and humidity is below 0.8 (the lightest areas on the map). This is probably because the weather conditions are ideal for outdoor activities.

We can also observe an intermediate point: if the temperature is between 0.25 and 0.4, rentals decrease to values between 3000 and 4000. The same happens if the temperature is favorable but humidity rises above 0.8; in that case, rentals also decline.

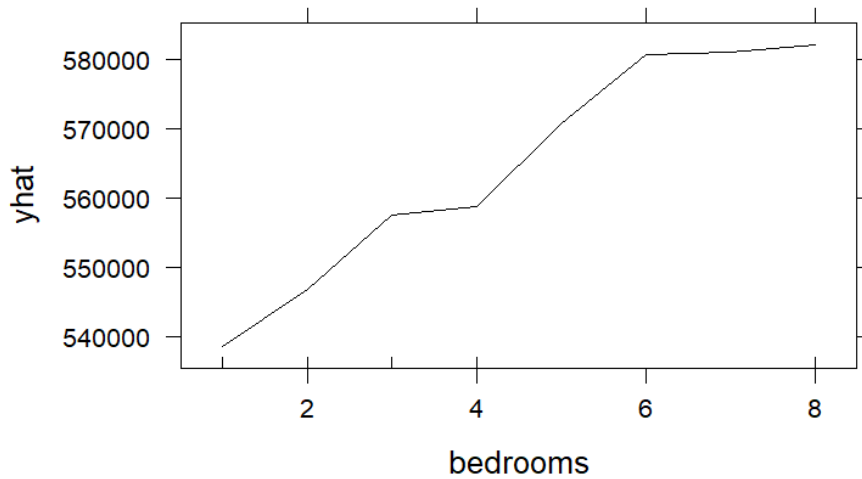
The worst-case scenario occurs when the lowest temperatures coincide with the highest humidity levels, where rentals barely reach 3000.

Looking at this graph as a whole, we realize that temperature is much more influential than humidity, since temperature defines the overall trend, while humidity only slightly reduces rentals in extreme cases (when it is very high).

QUESTION: Analyse the influence of bedrooms, bathrooms, sqft_living and floors on the predicted price.

Ahora analizaremos las distintas variables que provocan que una casa cueste más.

Influence of bedrooms

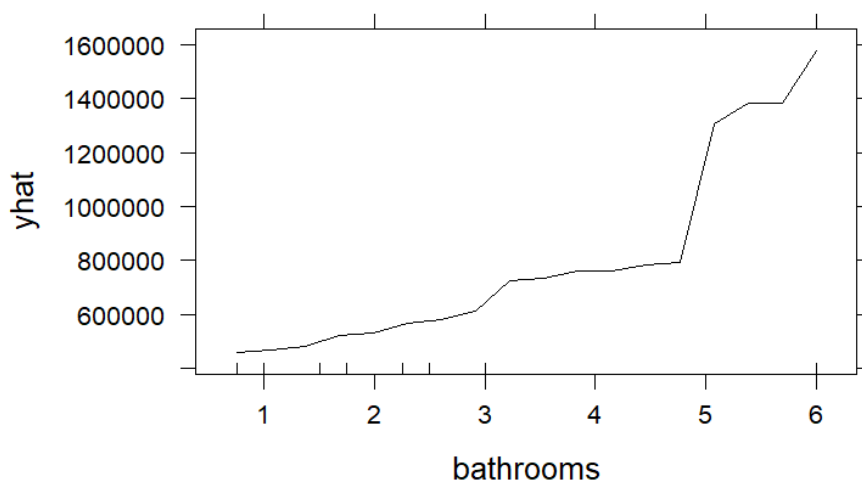


We observe that the number of bedrooms has an overall positive effect on price. However, there are points that seem contradictory, such as between 3 and 4 bedrooms or between 6 and 8 bedrooms, where the curve does not increase as sharply.

One possible explanation is that many older houses were built with 3 or 4 bedrooms, so even though they have more rooms, the age of the property may limit the increase in price.

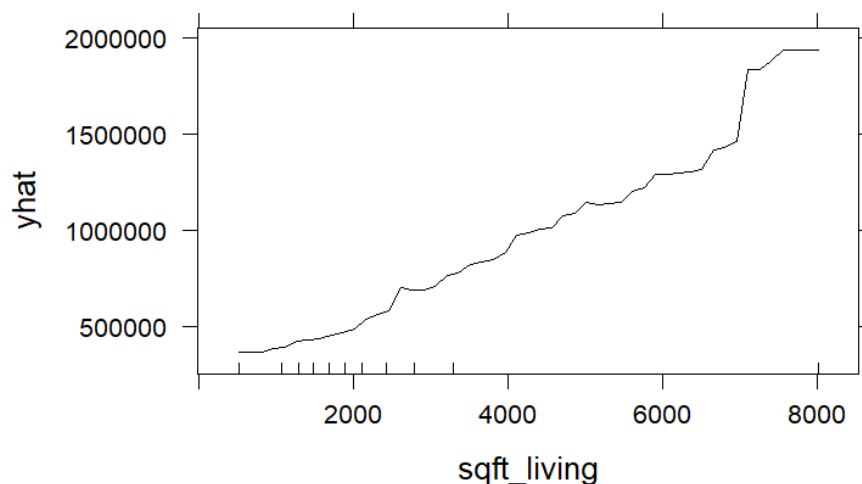
In the case of houses with 6 to 8 bedrooms, the fact that the price stabilizes around \$580,000 may be due to a limitation of the dataset itself (a small number of data points in that range), which prevents the prediction from clearly exceeding that threshold.

Influence of bathrooms



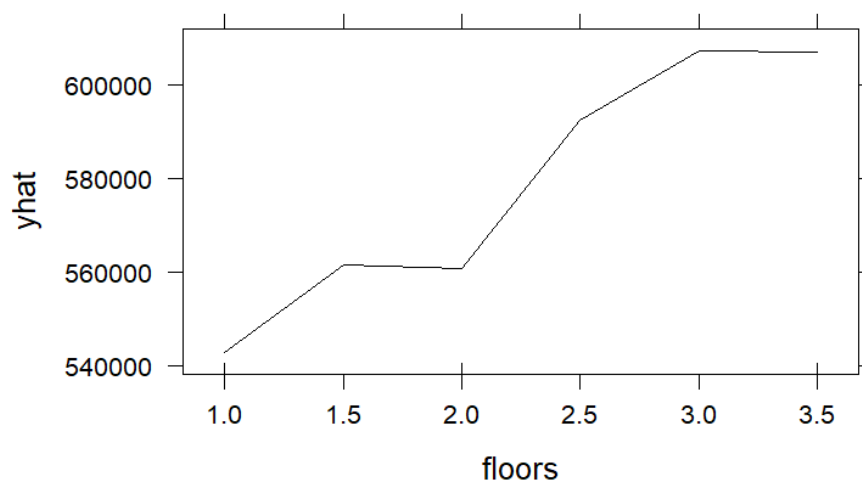
In this graph, we can see that for houses with between 1 and 4 bathrooms, the price remains within a similar range (between \$300,000 and \$600,000). However, when the house reaches 5 bathrooms, the price increases dramatically, reaching up to \$1,600,000. This is because mansions are typically the properties that have such a high number of bathrooms, while a conventional home rarely exceeds 3. The model identifies having 5 or more bathrooms as a direct indicator of a luxury property.

Influence of sqft_living



This is, without a doubt, the most influential variable in the model, with a price range varying between \$250,000 and \$2,000,000. We observe an almost linear increase in price: the larger the house, the more expensive it becomes. The small fluctuations or curves in the line indicate that size is not the only factor that matters; there are other external factors that the dataset does not specify (such as the exact location or neighborhood) that also affect the final value.

Influence of floors



This variable appears to be less significant than the previous ones. Moving from 1 to 2 floors represents an increase of around \$200,000, and from 2 to 3 floors the price rises by another \$400,000. However, once we reach the third floor, the price stabilizes. This suggests that buyers value usable space (square footage) or the number of rooms more than the fact that the house has multiple levels.